Positive-Part Matrix Trace Comparison, Quantitative Rigidity, and Ram Representations for the Matrix Lieb–Thirring Problem

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Abstract

The pinned matrix Lieb–Thirring manuscript reduces its sharp spectral estimate to an integrated constant-coefficient matrix resolvent comparison. We audit the full source dependency chain and give self-contained proofs of local extensions at that comparison, without claiming independent verification of the global spectral theorem. The matrix N in the comparison need not be assumed positive: its positive part satisfies an exact nonnegative deficit identity. Equality is equivalent to commutation of the two coefficient matrices under the source's uniform symbol coercivity hypothesis. A finite spectral-parameter window gives an explicit lower bound on this deficit in terms of the squared Hilbert–Schmidt norm of their commutator. We also prove continuum coercivity tests, improper-integral truncation and midpoint quadrature bounds, and residual-to-deficit error estimates. The private RamCS and physical-Gram representations supply precise coordinate and verification conventions, not an alternative sharp-constant theorem. Classical Fourier, variational, Riccati, and matrix-duality ingredients remain explicitly attributed, and no historical priority or computational speedup is asserted. Contribution assessment: C3/5, the positive-part extension with equality characterization and quantitative commutator control is a notable local refinement with explicit verification bounds but does not independently establish the full spectral theorem or its global stability.

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1 Pinned problem and scope

1.1 Exact spectral question

The source is OpenAI, *Sharp one-dimensional Lieb–Thirring inequalities for matrix potentials*, dated October 5, 2026, catalogue item 526, at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its local directory relative to the study's source root is

`preprints/Sharp-one-dimensional-Lieb-Thirring-inequalities-for-matrix-potentials-October-5-2026`.

We denote this source by [1]. The problem is on the whole real line, not a periodic box or a half-line. For every finite $m \geq 1$, every $1/2 < \gamma < 3/2$, and every measurable Hermitian $W : \mathbb{R} \to \mathbb{C}^{m\times m}$ such that

$$W(x) \geq 0 \text{ a.e.}, \qquad \int_{\mathbb{R}} \operatorname{tr} W(x)^{\gamma+1/2}\,\mathrm{d}x < \infty, \tag{1.1}$$

the source asserts

$$\operatorname{Tr}(H_W)_-^{\gamma} \leq C_\gamma \int_{\mathbb{R}} \operatorname{tr} W^{\gamma+1/2}\,\mathrm{d}x, \quad C_\gamma = \left(\frac{\gamma-1/2}{\gamma+1/2}\right)^{\gamma-1/2} \frac{\Gamma(\gamma+1)}{\sqrt{\pi}\Gamma(\gamma+3/2)}. \tag{1.2}$$

Here H_W is the self-adjoint operator associated with the closed semibounded form

$$h_W[u] = \int_{\mathbb{R}} \|u'\|^2\,\mathrm{d}x - \int_{\mathbb{R}} u^* W u\,\mathrm{d}x, \qquad D(h_W) = H^1(\mathbb{R};\mathbb{C}^m).$$

No H^2 operator-domain assertion is made for such a general measurable potential. The trace on the left counts all negative eigenvalues with multiplicity; the finite matrix trace on the right is unnormalized. Finite m is arbitrary, but no infinite-dimensional channel extension is proved here. The form domain specifies the boundary behavior: there are no finite-endpoint boundary conditions on $\mathbb{R}$.

The source supplies the sharpness example

$$W(x) = \operatorname{diag}((r+1)\operatorname{sech}^2(rx), 0, \dots, 0), \qquad r = (\gamma-1/2)^{-1} > 1.$$

The classification of all equality potentials is a different problem. In particular, the equality theorem in Section 5 concerns an auxiliary pair of *constant matrices*; it does not classify equality in (1.2).

1.2 Local problem and conventions

Set

$$0 < \sigma = \gamma - \tfrac{1}{2} < 1, \quad p = 1+\sigma, \quad q = 1+\sigma^{-1}, \quad \beta = \sigma - \tfrac{1}{2}. \tag{1.3}$$

Throughout the local analysis $B, V \in \mathbb{C}^{n\times n}$ are Hermitian and

$$X_s = (sI - B)^2 + V \geq bI \qquad \text{for every } s \in \mathbb{R}, \qquad b > 0. \tag{1.4}$$

The dimension n here is the number of trial rows in the action argument and need not equal the channel dimension m. We write $Y_s = (sI - B)^2$. The perturbation V is not the spatial potential W and is allowed to be indefinite. All norms without a subscript are operator norms; $\|A\|_{\mathrm{HS}}^2 = \operatorname{tr}(A^*A)$.

Inner products are linear in the second variable. For a Hermitian matrix A, A_+ is obtained by replacing its negative eigenvalues by zero.

Define

$$c_\sigma^{-1} = \int_0^\infty t^\beta (t^{-1/2} - (t+1)^{-1/2})\,\mathrm{d}t, \tag{1.5}$$

$$R(t) = \pi^{-1} \int_{\mathbb{R}} (X_s + tI)^{-1}\,\mathrm{d}s, \qquad P(t) = t^{-1/2} I - R(t), \tag{1.6}$$

$$N = c_\sigma \int_0^\infty t^\beta P(t)\,\mathrm{d}t. \tag{1.7}$$

The s and t integrals are matrix-norm integrals. In particular, (1.7) is an integral of the *difference*; the two separated t integrals generally diverge. Scaling in (1.5) gives

$$c_\sigma \int_0^\infty t^\beta (t^{-1/2} - (t+y)^{-1/2})\,\mathrm{d}t = y^\sigma \quad (y \geq 0). \tag{1.8}$$

For reference, Tonelli applied to $t^{-1/2} - (t+1)^{-1/2} = \frac{1}{2}\int_0^1 (t+y)^{-3/2}\,\mathrm{d}y$ also gives

$$c_\sigma = \frac{2\sigma}{\mathrm{B}(\sigma + 1/2, 1-\sigma)}. \tag{1.9}$$

Indeed the substitution $t = yu$ leaves $y^{\sigma-1}$, whose integral is $1/\sigma$. Both beta-function arguments are positive.

2 Stepwise source dependency audit

All seven substantive section files, the included historical note, the preamble, the main file, and the bibliography were read. The main source contains no mathematical appendix and includes no further mathematical section beyond these files. This is a source-reading audit, not a machine-checked or independent global proof certification. The following trace explicitly separates local reproduction from inherited global steps.

2.1 Forms, action, and the matrix comparison

In `sections/introduction.tex`, `thm:main` has precisely the quantifiers in Section 1. In `sections/spectral.tex`, `lem:form-compactness` uses $w = \|W\| \in L^p$ and

$$\int u^* W u \leq 2^{1/p} \|w\|_p \|u'\|_2^{1/p} \|u\|_2^{2-1/p} \leq \epsilon \|u'\|_2^2 + C_\epsilon \|u\|_2^2 .$$

Truncation in space and amplitude makes $W^{1/2} : H^1 \to L^2$ compact by the bounded-interval Rellich embedding. An orthonormal sequence in a negative spectral subspace bounded away from zero would contradict that compactness. Thus the form and the discrete negative spectrum are established under the stated integrability, without assuming bounded W.

In `sections/action-framework.tex`, `thm:action` fixes a bounded interval $\Omega = (x_-, x_+)$, integers $n, d \geq 1$, and

$$U \in H_0^1(\Omega; \mathbb{R}^{n\times d}), \qquad \int_\Omega U U^\mathsf{T}\,\mathrm{d}x = I_n, \qquad K = \mathrm{diag}(k_1, \ldots, k_n), \quad k_i > 0.$$

For lower triangular C, the action is

$$\mathcal{E}(CU) = \int_\Omega \left(\|(CU)'\|_{\mathrm{HS}}^2 + \operatorname{tr}(K^2 CUU^{\mathsf{T}}C^{\mathsf{T}}) - \operatorname{tr}((CUU^{\mathsf{T}}C^{\mathsf{T}})^q) \right) \mathrm{d}x.$$

Its asserted lower bound is $\sup_C \mathcal{E}(CU) \geq \sum_i \int_{-k_i}^{k_i} (k_i^2 - s^2)^\sigma \, \mathrm{d}s$. The interval is auxiliary and the trace-zero condition is indispensable when integrating the action identity.

In `sections/trace.tex`, `thm:trace` assumes (1.4) and $N \geq 0$ and proves $\operatorname{tr}(NV) \geq \operatorname{tr} N^q$. The chain is resolvent convergence, unique half-line solutions, Hermitian boundary maps, the exact gluing normalization $R^{-1} = (S+T)/2$, Riccati identities, a logarithmic determinant integral, weighted integration by parts, and a Hilbert–Schmidt square. Sections 3–5 reproduce this local chain and extend its final step. Positivity at sampled s alone does not satisfy the theorem.

2.2 Regularized field and endpoint construction

In `sections/field.tex`, `lem:field-regularity` defines f_δ for $y \geq 0$ by an integral of resolvent differences, extends it tangentially for $y < 0$, and uses $f_\delta'(y) = a_\sigma (y+\delta)^{\beta-1}$ on $y \geq 0$. The field

$$M_\delta(B) = \lim_{r\to\infty} \int_{-r}^{r} \{ f_\delta(s^2 I - 2sB + K^2) - f_\delta(s^2) I \} \, \mathrm{d}s$$

has an odd leading tail $-2s f_\delta'(s^2) B$. The remaining tail is $O(|s|^{2\sigma-3})$, integrable since $\sigma < 1$. Symmetric cutoffs, not an unqualified absolute integral of each field term, are essential. The primitive satisfies $dJ_\delta(B)[H] = -\operatorname{tr}(M_\delta(B)H)$ and $J_\delta(0) = 0$.

The directional Jensen bound in `lem:field-properties` is

$$e^{\mathsf{T}} M_\delta(B) e \leq \mu_\delta(e^{\mathsf{T}} K^2 e - (e^{\mathsf{T}} B e)^2), \quad \mu_\delta(z) = (z+\delta)^\sigma - \delta^\sigma \quad (z \geq 0).$$

Therefore $M_\delta(B) \geq 0$ implies $Q_s = s^2 I - 2sB + K^2 \geq 0$ for every real s, but does *not* imply $K^2 - B^2 \geq 0$. In `prop:field-comparison`, take $V = K^2 - B^2 + \delta I$. A jointly integrable resolvent-difference majorant justifies Fubini and gives

$$N = M_\delta(B) + \delta^\sigma I, \qquad \operatorname{tr}(M_\delta(B)(K^2 - B^2)) \geq \operatorname{tr} M_\delta(B)^q - E_\delta,$$

where $E_\delta = n\{\delta \kappa^{2\sigma} + \delta^\sigma(\kappa^2 + \delta)\}$ and $\kappa = \max_i k_i$. The positive-part extension does not remove the need for the directional positivity argument: the symbol coercivity assumption still has to hold.

In `sections/paths.tex`, `prop:paths` seeks

$$\epsilon B' = M_\delta(B) - (CU)(CU)^{\mathsf{T}}, \quad B(x_-) = K, \quad B(x_+) = -K.$$

A compactly supported equivariant cutoff makes the initial-value field globally Lipschitz. Integration at an endpoint zero bounds CC^{T}. No row of C can vanish, because the corresponding scalar diagonal equation would remain at k_i and miss $-k_i$. Hence the row-sign group acts freely at all zeros. At the initial homotopy parameter the equation is $CC^{\mathsf{T}} = 2\epsilon K$ and its 2^n diagonal solutions form a single regular sign orbit. Smoothing with a fixed initial collar and equivariant perturbation leads, if terminal zeros were absent, to a compact one-manifold quotient with exactly one boundary point. Boundary parity excludes this. Confinement then removes the cutoff. These topological and continuation arguments are inherited dependencies, not replaced by Fourier transforms or the finite certificates below.

2.3 Penalty limits and the spectral passage

In `sections/action-limit.tex`, the identity obtained from $J_\delta(B) + \operatorname{tr}(A^\mathsf{T} B A)$ includes the term $-\epsilon^{-1} \left\| M_\delta(B) - AA^\mathsf{T} \right\|_{\mathrm{HS}}^2$. For fixed δ, all pairs $(A(x), B(x))$ lie in one compact finite-dimensional set. Any sequence violating the desired uniform bound would have residual tending to zero; at its limit, the field comparison applies. This proves the penalty limsup, not strong convergence of the entire paths. First $\epsilon \downarrow 0$ with δ fixed, then $\delta \downarrow 0$ in the explicit endpoint integral. No uniform-in-δ compactness is silently required.

In `sections/spectral.tex`, smooth compactly supported approximations to a finite eigenfunction family are normalized by their Gram inverse square root and rotated to diagonalize the form. Ordering $k_1 \geq \cdots \geq k_n$ and using lower triangular C gives

$$h_W[a_i] + k_i^2 \left\| a_i \right\|_2^2 = \sum_{j \leq i} C_{ij}^2 (k_i^2 - k_j^2) \leq 0.$$

The nonzero eigenvalues of AA^T and $A^\mathsf{T} A$ agree. Scalar maximization and spectral-measure Jensen give

$$\operatorname{tr}(WZ) - \operatorname{tr} Z^q \leq D_\sigma \operatorname{tr} W^p, \qquad D_\sigma = \frac{\sigma^\sigma}{(1+\sigma)^{1+\sigma}}.$$

Dividing by $\mathrm{B}(1/2, 1+\sigma)$ recovers exactly C_γ. Supremizing finite eigenvalue sums obtains the full moment. Realification doubles both the spectral trace and $\operatorname{tr} W^p$, so the complex inequality has the same constant. Finally the source factors the scalar example as $Q^*Q - 1$ with $Q = \partial_x + \tanh(rx)$; its partner $QQ^* - 1 = -\partial_x^2 + (r-1)\operatorname{sech}^2(rx) \geq 0$ excludes other negative eigenvalues. These deductions identify how the sharp coefficient is used; this report does not announce independent verification of the endpoint-map proof or of the entire theorem.

3 RamCS realization and boundary normalization

3.1 Exact transform dictionary

The actual private passage [5], `eq:ramcs-definition`, uses the positive analysis kernel

$$\mathcal{R}_c f(a) = \int_{\mathbb{R}} f(x) e^{a(x-c)} \, \mathrm{d}x.$$

Its inversion theorem and Plancherel identity are weighted Fourier statements. On the unweighted contour used here, set $a = -is$ and

$$\mathcal{U}_c f(s) = (2\pi)^{-1/2} \mathcal{R}_c f(-is) = e^{isc} (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-isx} f(x) \, \mathrm{d}x.$$

Componentwise extension is unitary from $L^2(\mathbb{R}; \mathbb{C}^n)$ to itself. Integration by parts first on Schwartz functions, followed by closure, gives $\mathcal{U}_c \partial_x \mathcal{U}_c^{-1} = is$. Thus for

$$\mathcal{H}_{B,V} = -(\partial_x - iB)^2 + V, \qquad D(\mathcal{H}_{B,V}) = H^2(\mathbb{R}; \mathbb{C}^n), \tag{3.1}$$

one has $\mathcal{U}_c \mathcal{H}_{B,V} \mathcal{U}_c^{-1} = X_s$. This H^2 statement is valid because B, V are constant finite matrices and the leading part is $-\partial_x^2$. The associated form domain is H^1.

The corpus labels `eq:ramcs-convolution` and `eq:variable-plane-wave` explain the precise boundary of this correspondence. A constant-coefficient inverse is a matrix convolution multiplier, and its plane waves are generalized eigenfunctions, not normalizable L^2 states. A varying potential

$W(x)$ is not diagonalized by this transform: multiplication becomes frequency coupling. A nonzero real contour weight changes the Hilbert norm and has not been introduced here. Consequently neither contour deformation nor scalar reciprocal-symbol division proves (1.2).

Proposition 3.1 (Full-line inverse and its trace)**.** *Under* (1.4)*,* $\mathcal{H}_{B,V}+t$ *is invertible for* $t\ge 0$ *and* $\|(\mathcal{H}_{B,V}+t)^{-1}\|_{L^2\to L^2}\le (b+t)^{-1}$*. For* $t>0$ *its continuous Green matrix is*

$$G_t(x)=\frac{1}{2\pi}\int_{\mathbb{R}} e^{isx}(X_s+tI)^{-1}\,\mathrm{d}s,\qquad G_t(0)=\tfrac{1}{2}R(t).$$

Proof. Unitary multiplication by $(X_s+tI)^{-1}$ proves the norm estimate. Quadratic growth in s makes that inverse matrix integrable, so Fourier inversion defines a continuous G_t and evaluating at zero gives the factor $1/2$. The kernel identity holds distributionally, and convolution on a dense class agrees with the bounded inverse, hence extends to L^2. □

3.2 Half-lines with prescribed traces

Write $\nabla=\partial_x-iB$. On each half-line prescribe $\psi(0)=v\in\mathbb{C}^n$ and require $\psi\in H^1$. The differential equation is

$$(-\nabla^2+V+tI)\psi=0. \tag{3.2}$$

Whole-line coercivity and quadratic growth give a lower bound by a positive multiple of $1+s^2$. Zero extension therefore makes the form coercive on the trace-zero half-line H^1 space. For any H^1 lifting of v, solving for its trace-zero correction by the Riesz theorem gives a unique solution. Constant coefficients imply $\psi\in H^2$; both ψ and ψ' tend to zero at the infinite end.

Define the boundary matrices by

$$\nabla\psi(0+)=-Sv,\qquad \nabla\psi(0-)=Tv. \tag{3.3}$$

Integration by parts gives respectively the half-line energies v^*Sv and v^*Tv; polarization makes S,T Hermitian. They need not individually be positive. Translation and uniqueness give the same first-order relations at every interior point. Substitution yields

$$V+tI=S^2+i[B,S]=T^2-i[B,T]. \tag{3.4}$$

Gluing the two solutions gives a continuous function with derivative jump $-(S+T)v$. Hence its distributional image under $-\nabla^2+V+t$ is $\delta_0(S+T)v$, and Proposition 3.1 gives

$$v=\tfrac{1}{2}R(t)(S+T)v,\qquad L:=\tfrac{1}{2}(S+T)=R(t)^{-1}>0,\quad Z:=\tfrac{1}{2}(S-T). \tag{3.5}$$

The point source is in $H^{-1}(\mathbb{R})$, not $L^2(\mathbb{R})$. Using a full-line L^2 residual bound directly on this delta source would be invalid. The matrix-symbol certificates in Section 6 avoid that error.

The half-line solutions and maps depend continuously on $t>0$. For completeness, on a compact t interval coercivity is uniform; the difference of the two variational equations bounds the trace-zero correction in H^1 by a constant times $|t-t'|$. Equation (3.2) then bounds the second-derivative difference in L^2, and the H^2 trace theorem gives continuity of S,T. This justifies the pointwise equality arguments below without selecting discontinuous eigenbases.

Adding and subtracting (3.4) gives

$$V+tI=L^2+Z^2+i[B,Z],\qquad LZ+ZL+i[B,L]=0. \tag{3.6}$$

In an eigenbasis of L, $Z_{ii}=0$. In particular $\operatorname{tr}Z=0$. Multiplication of the second relation by R on both sides gives $i[R,B]=-(ZR+RZ)$ and therefore

$$\operatorname{tr}(R(V+tI))=\operatorname{tr}L-\operatorname{tr}(RZ^2). \tag{3.7}$$

The correction is nonnegative because $\operatorname{tr}(RZ^2)=\operatorname{tr}(ZRZ)\ge 0$.

4 A positive-part trace extension

4.1 Convergence and the inherited weighted identity

For fixed matrices satisfying (1.4), quadratic growth gives $R(t) = O(1)$ as $t \downarrow 0$. Diagonalizing B gives $\pi^{-1}\int (Y_s + tI)^{-1}\,\mathrm{d}s = t^{-1/2}I$. For large t, the resolvent identity and a lower bound of the form $c(s^2+t)I$ for both symbols imply

$$\|P(t)\| = O(t^{-1/2})\ (t \downarrow 0), \qquad \|P(t)\| = O(t^{-3/2})\ (t \to \infty). \tag{4.1}$$

Thus N is well defined and Hermitian regardless of its sign. Fully explicit versions of these estimates appear in Section 6.

Define

$$F(t) = \frac{\operatorname{tr} V}{\sqrt{t}} - \frac{1}{\pi}\int_{\mathbb{R}} \log \frac{\det(X_s + tI)}{\det(Y_s + tI)}\,\mathrm{d}s, \tag{4.2}$$

$$G(t) = \operatorname{tr}(L - 2\sqrt{t}I + tR). \tag{4.3}$$

The determinant ratio is positive and its logarithm is $O(s^{-2})$. Differentiation in t is locally dominated by the resolvent identity. The first relation in (3.4) factors

$$X_s + tI = (sI - B + iS)(sI - B - iS).$$

Every eigenvalue $z = u + iv$ of $B + iS$ has $v > 0$: its eigenvector generates the right solution $e^{izx}v_0$, which must be in $L^2(0,\infty)$. Thus the determinant is the product of $(s-u_j)^2 + v_j^2$ with algebraic multiplicity. Symmetric cutoffs permit translating each center since the derivative of $\log(s^2+v^2)$ tends to zero at infinity. Integrating the centered scalar identity gives

$$\int_{\mathbb{R}} \log \frac{s^2+v^2}{s^2+w^2}\,\mathrm{d}s = 2\pi(v-w), \qquad v, w > 0.$$

Since $\sum v_j = \operatorname{tr} S = \operatorname{tr} L$, this proves

$$F(t) = \frac{\operatorname{tr} V}{\sqrt{t}} - 2\operatorname{tr}(L - \sqrt{t}I) = t^{-1/2}\operatorname{tr}((L - \sqrt{t}I)^2 + Z^2) \geq 0. \tag{4.4}$$

Also $G \geq 0$, since $r^{-1} - 2\sqrt{t} + tr = (1 - \sqrt{t}r)^2/r$ for every eigenvalue $r > 0$ of R. Equations (3.7) and (4.4) give

$$\operatorname{tr}(PV) = F + G + \operatorname{tr}(RZ^2). \tag{4.5}$$

In particular all three terms are bounded by $\operatorname{tr}(PV)$ despite possible indefiniteness of P, V. The bounds (4.1) show their weighted integrability and $t^{\beta+1}F(t) \to 0$ at both ends. Differentiating (4.2) and using (4.4) gives $tF' = -F/2 - G$. Integration by parts now yields

$$\int_0^\infty t^\beta G\,\mathrm{d}t = \sigma\int_0^\infty t^\beta F\,\mathrm{d}t, \quad \operatorname{tr}(NV) = qc_\sigma\int_0^\infty t^\beta G\,\mathrm{d}t + c_\sigma\int_0^\infty t^\beta \operatorname{tr}(RZ^2)\,\mathrm{d}t. \tag{4.6}$$

These are the source's weighted identities, with their convergence and sign requirements made explicit. None uses $N \geq 0$.

Theorem 4.1 (Positive-part comparison with exact deficit)**.** *Let* (1.4) *hold for Hermitian* B, V*, and define* (1.5)–(1.7)*. No sign assumption on* N *is required. With* $D = (N_+)^{1/\sigma}$ *and*

$$\mathcal{D}_D(t) = \left\| R(t)^{-1/2} - R(t)^{1/2}(tI + D)^{1/2} \right\|_{\mathrm{HS}}^2, \tag{4.7}$$

the absolutely convergent identity is

$$\Delta := \operatorname{tr}(NV) - \operatorname{tr}((N_+)^q) = c_\sigma\int_0^\infty t^\beta\{\operatorname{tr}(R(t)Z(t)^2) + q\mathcal{D}_D(t)\}\,\mathrm{d}t \geq 0. \tag{4.8}$$

Proof. For any fixed $D \geq 0$, put

$$F_0(t, D) = \frac{\operatorname{tr} D}{\sqrt{t}} - 2\operatorname{tr}((tI + D)^{1/2} - \sqrt{t}I).$$

Diagonalizing D and integrating the scalar identity

$$\frac{d}{\sqrt{t}} - 2(\sqrt{t+d} - \sqrt{t}) = \int_0^d (t^{-1/2} - (t+y)^{-1/2})\,\mathrm{d}y$$

with Tonelli and (1.8) gives $c_\sigma \int_0^\infty t^\beta F_0(t, D)\,\mathrm{d}t = p^{-1} \operatorname{tr} D^p$. Expanding the square (4.7) using cyclicity, not a commutation assumption, gives

$$\mathcal{D}_D = \operatorname{tr}(R^{-1} + R(tI + D) - 2(tI + D)^{1/2}) = G - \operatorname{tr}(PD) + F_0(t, D).$$

All three terms on the right are integrable by the bounds above. Therefore

$$c_\sigma \int_0^\infty t^\beta G\,\mathrm{d}t = \operatorname{tr}(ND) - \frac{1}{p}\operatorname{tr} D^p + c_\sigma \int_0^\infty t^\beta \mathcal{D}_D\,\mathrm{d}t. \tag{4.9}$$

For $D = (N_+)^{1/\sigma}$, functional calculus in a single eigenbasis of N gives $\operatorname{tr}(ND) = \operatorname{tr}((N_+)^q) = \operatorname{tr} D^p$ even when N has negative eigenvalues. Since $1 - 1/p = 1/q$, substitution in (4.6) proves (4.8). □

This is an extension of the source's hypothesis, not evidence that an admissible pair with indefinite N exists. No such witness is asserted. The extension is useful in a verification procedure because an approximate N need not first be certified positive before a deficit comparison can be formulated. Convex positive-part duality and Hilbert–Schmidt completion of squares are classical; their application to this specific source identity is the local refinement.

5 Equality and quantitative noncommutativity

Theorem 5.1 (Exact local equality criterion)**.** *Under the hypotheses of Theorem 4.1, $\Delta = 0$ if and only if $[B, V] = 0$. In that event $V \geq bI$ and $N = V^\sigma > 0$.*

Proof. The summands in (4.8) are continuous and nonnegative for $t > 0$. If their positive-weight integral is zero, they vanish at every t. Since $R > 0$, $\operatorname{tr}(RZ^2) = 0$ implies $Z = 0$. Equations (3.6) then yield $V + tI = L^2$ and $[B, L] = 0$, hence $[B, V] = 0$ at any one t. Simultaneous unitary diagonalization writes X_s with diagonal entries $(s - b_i)^2 + v_i$. Evaluating at $s = b_i$ gives $v_i \geq b$. The scalar resolvent integral and (1.8) give $R(t) = (tI + V)^{-1/2}$ and $N = V^\sigma$.

Conversely, commutation and (1.4) give these same formulas, so $\operatorname{tr}(NV) = \operatorname{tr} V^{1+\sigma} = \operatorname{tr} N^q$. Equivalently, the half-line maps both equal $(V + tI)^{1/2}$ and both squares in (4.8) vanish. This proves the equivalence. □

Proposition 5.2 (Exact commutator density)**.** *At fixed $t > 0$, diagonalize L with eigenvalues $\ell_i > 0$. In this basis*

$$Z_{ij} = i\frac{\ell_i - \ell_j}{\ell_i + \ell_j} B_{ij}, \quad Z_{ii} = 0, \tag{5.1}$$

and

$$\operatorname{tr}(RZ^2) = \sum_{i<j} \frac{(\ell_i - \ell_j)^2}{\ell_i \ell_j (\ell_i + \ell_j)} |B_{ij}|^2 \geq \frac{\|[B, L]\|_{\mathrm{HS}}^2}{4\,\|L\|^3}. \tag{5.2}$$

Proof. Take entries in the second equation of (3.6). Then sum $\operatorname{tr}(RZ^2) = \sum_{i,j} \ell_i^{-1} |Z_{ij}|^2$ in unordered pairs. The denominator in each resulting pair is at most $2\|L\|^3$, whereas $\|[B,L]\|_{\mathrm{HS}}^2 = 2\sum_{i<j}(\ell_i - \ell_j)^2 |B_{ij}|^2$. This proves both the exact formula and its bound. Repeated eigenvalues simply give zero terms, so no simplicity hypothesis is hidden here. □

Theorem 5.3 (A finite-window bound for the original coefficients)**.** *Fix $0 < a < d < \infty$, put $M = \|B\|$, $v = \|V\|$, and define*

$$H = \sqrt{n(v+d)}, \qquad K = H^2 + MH + M^2, \qquad J_\beta(a,d) = \frac{d^{\beta+1} - a^{\beta+1}}{\beta+1}. \tag{5.3}$$

Then

$$\Delta \geq \frac{c_\sigma J_\beta(a,d)}{16HK^2} \|[B,V]\|_{\mathrm{HS}}^2 . \tag{5.4}$$

In particular a noncommuting admissible pair has strictly positive deficit, and on fixed bounded coefficient sets the deficit controls the commutator quantitatively.

Proof. Taking the trace of (3.6) gives $\operatorname{tr} L^2 + \operatorname{tr} Z^2 = \operatorname{tr} V + nt \leq n(v+t)$. Thus for $t \in [a,d]$, $\|L\|, \|Z\| \leq H$. The second equation gives $\|[B,L]\|_{\mathrm{HS}} \leq 2\|L\| \, \|Z\|_{\mathrm{HS}}$. Commuting the first equation with B and using submultiplicativity gives

$$\begin{aligned} \|[B,V]\|_{\mathrm{HS}} &\leq \left\|[B,L^2]\right\|_{\mathrm{HS}} + \left\|[B,Z^2]\right\|_{\mathrm{HS}} + \|[B,i[B,Z]]\|_{\mathrm{HS}} \\ &\leq 4\|L\|^2 \|Z\|_{\mathrm{HS}} + 4M\|Z\| \, \|Z\|_{\mathrm{HS}} + 4M^2 \|Z\|_{\mathrm{HS}} \leq 4K \|Z\|_{\mathrm{HS}} . \end{aligned}$$

Also $R = L^{-1} \geq H^{-1} I$, so $\operatorname{tr}(RZ^2) \geq H^{-1}\|Z\|_{\mathrm{HS}}^2 \geq \|[B,V]\|_{\mathrm{HS}}^2 / (16HK^2)$. Integrate only over $[a,d]$ in (4.8) and discard the other nonnegative terms. Here H, K are strictly positive since $d > 0$. □

The coefficient is deliberately conservative and depends on matrix size and coefficient bounds. It is not a uniform stability theorem as $n \to \infty$, nor a norm-distance estimate to a globally commuting pair. It also does not control the deficit of the spatial Lieb–Thirring inequality; transporting it through the penalized paths would require further estimates.

Proposition 5.4 (Resolvent-shape control)**.** *For $D = (N_+)^{1/\sigma}$ and every $t > 0$,*

$$\mathcal{D}_D(t) \geq \frac{t}{\|R(t)\|} \left\| R(t) - (tI + D)^{-1/2} \right\|_{\mathrm{HS}}^2 . \tag{5.5}$$

Consequently (4.8) bounds the weighted integral of this shape error on every finite window.

Proof. Let $A = (tI + D)^{1/2}$ and $E = R^{-1/2} - R^{1/2} A$. Then $R - A^{-1} = -R^{1/2} E A^{-1}$. Since $\|A^{-1}\| \leq t^{-1/2}$, taking Hilbert–Schmidt norms and squaring proves the claim. No commutation of R and D is used. □

6 Continuum and numerical certificates

The following estimates make a finite approximation accountable to the actual improper integral. They prove mathematical error bounds; ordinary floating-point evaluations alone do not instantiate a rigorous interval certificate.

6.1 Testing the entire real symbol line

Proposition 6.1 (Directional and finite-mesh tests)**.** *For Hermitian B, V, condition (1.4) is equivalent to*

$$e^*(B^2+V)e-(e^*Be)^2\geq b \quad \text{for every unit } e\in\mathbb{C}^n. \tag{6.1}$$

Alternatively let $S>M=\|B\|$ obey $(S-M)^2-v\geq b$, where $v=\|V\|$. On $[-S,S]$ use a mesh containing its endpoints with maximum spacing h. If at every mesh point s_j a valid lower eigenvalue bound is

$$\lambda_{\min}(X_{s_j})\geq b+(S+M)h, \tag{6.2}$$

then (1.4) holds for all real s.

Proof. For a unit e, complete the square:

$$e^*X_se=(s-e^*Be)^2+e^*(B^2+V)e-(e^*Be)^2.$$

Minimizing over s proves the equivalence, with the order of the universal quantifiers unchanged. For $|s|\geq S>M$, $X_s\geq((|s|-M)^2-v)I\geq bI$. On the compact interval $\|X_s'\|\leq 2(S+M)$. The nearest mesh point is within $h/2$, so the perturbation is at most $(S+M)h$. The Rayleigh quotient minimum gives the desired lower bound between nodes. □

Remark 6.2 (An exact indefinite perturbation). Take

$$B=\begin{pmatrix}2&0\\0&-2\end{pmatrix},\qquad V=\begin{pmatrix}1&3/2\\3/2&1\end{pmatrix}.$$

The eigenvalues of V are $5/2,-1/2$, but

$$\lambda_{\min}(X_s)=s^2+5-\sqrt{16s^2+9/4}\geq 55/64.$$

Indeed minimizing in $y=s^2\geq 0$ gives $y=247/64$ and value $55/64$. Thus indefiniteness of V is genuinely allowed, not merely formal. This example makes no assertion about the sign of its integrated N.

6.2 Explicit improper-integral tails

Let $M=\|B\|$, $v=\|V\|$ and fix a valid coercivity margin $b>0$. Define

$$\alpha=\frac{b}{4(b+M^2+v)},\quad C_0=\frac{1}{\sqrt{\alpha b/2}},\quad C_\infty=\frac{\sqrt{2}v}{\sqrt{\alpha}},\quad T_0=\max\{1,2M^2\}. \tag{6.3}$$

Lemma 6.3 (Explicit resolvent estimates)**.** *For all real s and $t\geq 0$,*

$$X_s+tI\geq(\alpha s^2+b/2+t)I,\qquad \|R(t)\|\leq\frac{1}{\sqrt{\alpha}\sqrt{b/2+t}}. \tag{6.4}$$

For $t>0$, $\|P\|(t)\leq t^{-1/2}+C_0$; for $t\geq T_0$, $\|P\|(t)\leq C_\infty t^{-3/2}$.

Proof. The elementary inequality $(s-\lambda)^2\geq s^2/2-\lambda^2$ gives $X_s\geq(s^2/2-M^2-v)I$. Combine this lower bound with bI, using weight $b/[2(b+M^2+v)]$ for the first. This proves the first part of (6.4); scalar integration proves the second and the small-t bound. For $t\geq T_0$, $Y_s+tI\geq(s^2/2+t/2)I\geq(\alpha s^2+t/2)I$, and the same weaker lower bound holds for X_s+tI. The resolvent identity then gives

$$\|P\|(t)\leq\frac{v}{\pi}\int_{\mathbb{R}}(\alpha s^2+t/2)^{-2}\,\mathrm{d}s=\frac{\sqrt{2}v}{\sqrt{\alpha}}t^{-3/2}.$$

□

Theorem 6.4 (A complete two-cutoff error bound)**.** *Let $0 < \epsilon < T$, $T \geq T_0$, and $S > 0$. Set*

$$N_{\epsilon,T,S} = c_\sigma \left[\frac{T^\sigma - \epsilon^\sigma}{\sigma} I - \frac{1}{\pi} \int_\epsilon^T \int_{-S}^S t^\beta (X_s + tI)^{-1} \, \mathrm{d}s \, \mathrm{d}t \right]. \tag{6.5}$$

Then $\|N - N_{\epsilon,T,S}\| \leq E_{\mathrm{tail}}$, where

$$E_{\mathrm{tail}} = c_\sigma \left[\frac{\epsilon^\sigma}{\sigma} + \frac{C_0 \epsilon^{\sigma+1/2}}{\sigma + 1/2} + \frac{C_\infty T^{\sigma-1}}{1-\sigma} + \frac{2J_\beta(\epsilon, T)}{\pi \alpha S} \right]. \tag{6.6}$$

Proof. Integrate the small- and large-t estimates in Lemma 6.3 against $c_\sigma t^\beta$. For the remaining s tail use $\|(X_s + tI)^{-1}\| \leq 1/(\alpha s^2)$, whose two-sided integral outside $[-S, S]$ is $2/(\alpha S)$. Integrate this bound over $[\epsilon, T]$. The scalar part of (6.5) is exact, since $\beta - 1/2 = \sigma - 1$. □

The last bound is intentionally simple and can be expensive when T is large. Integrating the difference of the two resolvents instead can improve the spatial tail rate after stronger simultaneous exterior bounds are proved. Such an improvement is not presumed in (6.6).

6.3 Midpoint rules and solve residuals

Partition $[-S, S] \times [\epsilon, T]$ into rectangles with maximum side lengths h_s, h_t. Define

$$w_0 = \max\{\epsilon^\beta, T^\beta\}, \qquad w_1 = \max\{\epsilon^{\beta-1}, T^{\beta-1}\},$$
$$L_s = \frac{2w_0(S+M)}{(b+\epsilon)^2}, \qquad L_t = \frac{|\beta| w_1}{b+\epsilon} + \frac{w_0}{(b+\epsilon)^2}.$$

Differentiating $F(s,t) = t^\beta (X_s + tI)^{-1}$ proves $\|\partial_s F\| \leq L_s$ and $\|\partial_t F\| \leq L_t$ on this rectangle. Thus its midpoint sum Q satisfies

$$\left\| Q - \int_\epsilon^T \int_{-S}^S F(s,t) \, \mathrm{d}s \, \mathrm{d}t \right\| \leq \frac{2S(T-\epsilon)}{4} (L_s h_s + L_t h_t). \tag{6.7}$$

To see the constant, on each rectangle use the straight coordinate paths from its midpoint. The average absolute displacement along a coordinate of width h is $h/4$. Summing the integrals of these bounds gives (6.7).

If Z_{jk} approximates $(X_{s_j} + t_k I)^{-1}$, its full matrix residual $E_{jk} = I - (X_{s_j} + t_k I) Z_{jk}$ gives

$$\left\| Z_{jk} - (X_{s_j} + t_k I)^{-1} \right\| \leq \frac{\|E_{jk}\|}{b + t_k}. \tag{6.8}$$

Multiply each such bound by its nonnegative quadrature weight t_k^β and its cell area, then by c_σ/π, and add to the cutoff and midpoint bounds. Bounds for input error, scalar normalization, arithmetic, matrix norms and eigenvalues must also be added for a validated computation. Symmetrizing the final approximate matrix does not increase its operator-norm error relative to Hermitian N.

Proposition 6.5 (From an N error to a deficit error)**.** *Suppose a Hermitian $\widehat{N}$ satisfies $\left\| N - \widehat{N} \right\| \leq \eta$, and set $K_N = \left\| \widehat{N} \right\| + \eta$. With B, V exact, define $\widehat{\Delta} = \mathrm{tr}(\widehat{N} V) - \mathrm{tr}((\widehat{N}_+)^q)$. Then*

$$|\Delta - \widehat{\Delta}| \leq n(v + q K_N^{q-1}) \eta. \tag{6.9}$$

Proof. The linear trace difference is at most $nv\eta$. Ordered Hermitian eigenvalues differ by at most η by the min–max principle. On $[-K_N, K_N]$, the scalar function $x \mapsto (x_+)^q$ is Lipschitz with constant qK_N^{q-1}. Apply this scalar bound to each ordered eigenvalue and sum. This proof does not assume an operator-Lipschitz theorem for noninteger matrix powers. □

These estimates provide a terminating finite procedure for an arbitrarily prescribed positive error tolerance, provided the inputs and the coercivity margin can themselves be certified: first shrink ϵ, then enlarge T, then enlarge S, then refine both midpoint meshes and solve tolerances. The bounds are not efficient complexity estimates. A finite calculation verifies only its specified pair and parameter range, never the source's universal theorem.

7 Physical metrics and checked matrix operations

7.1 Exact finite coordinates versus a compressed operator

The relevant private physical-metric passage [6] defines injective synthesis $J : \mathbb{C}^k \to H$, $G = J^*J$, $P = JG^{-1}J^*$, and $K_J = J^*AJ$. The coefficient operator is $B_J = G^{-1}K_J$. If A is symmetric, then $GB_J = B_J^*G$, not necessarily $B_J = B_J^*$. The full residual identity is

$$\|(A-\lambda)Jc\|_H^2 = \|(B_J-\lambda)c\|_G^2 + \|(I-P)AJc\|_H^2 . \tag{7.1}$$

These are exactly `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, and `v21c01:residual-identity`, not conclusions inferred from a routing map.

For the constant n-channel model, an invertible channel change $T : \mathbb{C}^n \to \mathbb{C}^n$ is exact synthesis. Put $G = T^*T$, $B_G = T^{-1}BT$, $V_G = T^{-1}VT$. Then

$$X_{G,s} = (sI - B_G)^2 + V_G = T^{-1}X_sT, \quad GX_{G,s} = X_{G,s}^*G, \quad z^*GX_{G,s}z \geq bz^*Gz.$$

The trace, matrix functional calculus, and integrated comparison are transported by similarity, but all positive parts and adjoints are interpreted in the G metric or after whitening by $G^{1/2}$. Taking an ordinary Euclidean positive part of the generally non-Hermitian entries of $T^{-1}NT$ is not legitimate.

The exact node solve error obeys

$$\|z - \widehat{z}\|_G \leq (b+t)^{-1} \|f - (X_{G,s} + tI)\widehat{z}\|_G .$$

This follows by applying T and the Hermitian inverse bound. If only $aG \leq \widehat{G} \leq dG$ is certified, the corresponding computable estimate is

$$\|z - \widehat{z}\|_G \leq (b+t)^{-1}a^{-1/2} \|r\|_{\widehat{G}}, \quad \|z - \widehat{z}\|_{\widehat{G}} \leq \sqrt{d/a}(b+t)^{-1} \|r\|_{\widehat{G}} .$$

This uses the actual metric-equivalence statement `v21c01:gram-equivalence`; entrywise small Gram errors alone do not imply relative control.

For a genuine finite-dimensional spatial approximation, J is not surjective and the omitted residual in (7.1) remains. With $A = \mathcal{H}_{B,V}$, its columns must lie in H^2 for this L^2 residual formulation. A projected zero residual neither proves a continuum eigenvalue nor bounds the full Green trace. Boundary lift and quadrature errors are distinct from algebraic solve residuals.

7.2 Schur, Woodbury, and polynomial correspondences

The actual passage [7], `prop:schurresidual` and `eq:schurresidual`, states for a partitioned node matrix $A = X_s + tI$ that

$$r_I = f_I - A_{IB}\widehat{x}_B - A_{II}\widehat{x}_I, \quad r_S = f_B - A_{BI}A_{II}^{-1}f_I - (A_{BB} - A_{BI}A_{II}^{-1}A_{IB})\widehat{x}_B,$$

and $r_B = r_S + A_{BI}A_{II}^{-1}r_I$. Direct substitution proves the identity. Here the actual matrix is positive definite, so its principal block and Schur complement are positive definite. An interface-only residual omits the transported interior defect. Using the complete (r_B, r_I) in (6.8) is the source-specific correspondence.

For a fixed B and low-rank change $V \mapsto V + UV_0^*$, the private `thm:woodbury` is precisely the classical Sherman–Morrison–Woodbury identity

$$(A + UV_0^*)^{-1} = A^{-1} - A^{-1}U(I + V_0^*A^{-1}U)^{-1}V_0^*A^{-1}.$$

It applies when both the base inverse and the displayed middle inverse exist. The full updated symbol must still have a certified lower margin; a small middle-matrix condition number alone does not establish it. Updates in B also change $B^2 - 2sB$ and must include that entire change. No low-rank reuse timing or speedup has been measured in this report.

The same private report's subsection *Polynomial evaluation without eigendecomposition*, especially `eq:filtererror`, gives the spectral-theorem bound for a scalar uniform polynomial approximation. At a node, if $\operatorname{spec}(X_s + tI) \subset [\ell, u]$ with $\ell > 0$ and $\sup_{z\in[\ell,u]} |z^{-1} - p_k(z)| \le \epsilon_k$, then $\|(X_s + tI)^{-1} - p_k(X_s + tI)\| \le \epsilon_k$. This substitutes a known approximation error in the same quadrature ledger. A sampled scalar approximation error is not a uniform bound without derivative or interval control. The polynomial estimate, Schur elimination and Woodbury algebra are classical techniques; the private Ram contribution here is their organized use with physical norms, source terms and full residuals.

8 Verification, comparisons, and limits

8.1 Proof-level and finite checks

The positive-part extension is proved by scalar functional calculus in an eigenbasis of N, so zero or negative eigenvalues cause no undefined powers. The equality proof uses continuity of the half-line maps rather than an unproved choice of continuous eigenvectors. The commutator identities handle multiplicities directly. The cutoff proof controls both improper integrations; the midpoint proof controls the gaps between sample nodes. These are analytic verifications, independent of any numerical sample.

The executed finite verification fixtures are the diagonal commuting reduction, the exact indefinite-V example of Remark 6.2, Hermitian 2×2 Riccati and trace identities, and nonorthogonal-coordinate metric symmetry. They are deliberately bounded and do not search a continuum or infer universal positivity from random samples. Numerical residuals are checks in floating-point arithmetic unless accompanied by outward-rounded bounds. Neither finite fixtures nor a successful LaTeX build verifies the global Lieb–Thirring theorem.

For $B = \operatorname{diag}(0.3, -0.4)$, $V = \operatorname{diag}(1, 2)$, and $\sigma \in \{1/4, 1/2, 3/4\}$, finite Gauss rules with $(96, 192)$ and $(192, 384)$ nodes in the two integrations were compared with $N = V^\sigma$. The largest error at the finer resolution was 1.27×10^{-14}. The integration maps were $t = \tan^4\theta$ and $s = \sqrt{t + \|B\|^2 + \|V\| + 1}\tan\phi$, on finite angular intervals. For $t < 1$, the free resolvent was integrated

analytically; for $t \geq 1$, the resolvent difference was evaluated directly. A preceding test integrating both resolvents numerically failed at small t: the $\sigma = 1/4$ normalization error was about 0.0763. This failure illustrates why apparent finite quadrature convergence is not an automatic certificate.

For the exact indefinite-V fixture and $\sigma = 1/2$, the corrected finer rule gave

$$N \approx \begin{pmatrix} 0.9406635974 & 0.4241863907 \\ 0.4241863907 & 0.9406635974 \end{pmatrix}, \quad \Delta \approx 0.4736530413.$$

Its two eigenvalues were approximately 0.5164772 and 1.3648500, so this is not an indefinite-N witness. The two resolutions differed in operator norm by 1.53×10^{-14}. The analytic lower bound (5.4), using $[a, d] = [1, 2]$, is approximately 0.00207756. At $t = 1/2, 1, 2$, a 4×4 first-order system constructed the decaying half-line subspaces. The largest right-Riccati residual was 1.28×10^{-14}, and the largest discrepancy in the pairwise formula (5.2) was 4.17×10^{-16}. These values check implementation consistency, not rigorous decimal enclosures or the full integral identity.

A separately inspected exact symbolic test record contains 38 passing finite checks: three prescribed positive diagonal L fixtures, their Sylvester and Riccati identities, trace identities and pairwise remainders, a noncommuting-D residual-square expansion, and the positive-part optimizer on a prescribed indefinite Hermitian matrix. The last fixture tests algebra only, not realization of that matrix as an integrated N. In the fixed-t fixtures the matrix $V + tI$ is constructed from L, Z, B; they are not a certified whole-t family for a fixed V. This distinction prevents finite algebraic tests from being mistaken for the analytic half-line construction.

8.2 Public mathematical context as of the report date

The sharp matrix $\gamma = 3/2$ estimate belongs to the established work of Laptev–Weidl [2]; Benguria–Loss [3] give its Riccati/commutation proof. The present boundary maps and factorization should therefore not be described as discoveries of Riccati factorization or Fourier diagonalization. Read–Schulz, version 2 dated September 14, 2026, state the sharp exponent-one result and its operator-valued extension in Proposition 4 and equation (2.1) [4]. Their constant is $4/(3\sqrt{3}\pi)$, exactly (1.2) at $\gamma = 1$. That case is not a new contribution of this report or of the source's general-exponent assertion. The public sources were consulted by public topic and title; the confidential corpus is used only locally.

The source's finite-interval continuation proof covers arbitrary $0 < \sigma < 1$ and arbitrary finite channel dimension as a claimed theorem. The present contribution is different: it strengthens a constant-matrix trace statement, quantifies its equality obstruction, and supplies explicit finite verification bounds. It neither reproduces the entire scalar companion nor proves the source's topological endpoint argument independently.

8.3 Advantages and costs

The positive-part statement avoids an unnecessary sign gate on N and still retains an exact nonnegative remainder. Equality now has a simple invariant criterion, $[B, V] = 0$. The commutator estimate identifies an actual quantity controlled by the local deficit, instead of merely noting that two squares are nonnegative. The physical-Gram account makes nonorthogonal representations safe to compare with the Hermitian source matrices. The truncation and residual estimates attach finite calculations to the correct continuum integral.

The costs are equally explicit. The global symbol margin is indispensable and can be difficult to certify. The quantitative constant worsens with n, coefficient size, and an unsuitable window. The elementary quadrature bounds can be very conservative. The algebraic positive-part extension

alone does not provide an indefinite-N example. The matrix pairs arising along the penalized field paths do not automatically obey uniform margins as $\delta \downarrow 0$. Thus local quantitative rigidity cannot be passed through those limits without additional work.

8.4 Future mathematical questions

A natural next problem is to decide which signs of N are realizable under (1.4), rather than assuming the formal sign extension is strict. Another is to optimize the finite-window commutator constant using the exact pairwise density (5.2) and spectral separation information. A third is to propagate local deficits through the regularized field without losing them in the penalty and δ limits. Such a result would require compactness and coercivity information not provided here and could lead toward spatial stability estimates. Validated adaptive integration of the resolvent difference, together with precise half-line boundary residuals in H^{-1}, is a computational direction; no validated implementation or speed claim is asserted.

9 Contribution assessment

The realized contribution is rated C3/5, a notable local refinement relative to the pinned source: Theorem 4.1 removes the sign hypothesis in a positive-part formulation, Theorem 5.1 identifies all equality pairs, and Theorem 5.3 proves explicit quantitative noncommutativity control. Theorems and propositions in Section 6 turn the source's asymptotic resolvent bounds into finite-input error estimates. These gains are specific to the source trace mechanism and go beyond a change of notation, but they are direct local consequences and estimates rather than a newly established sharp spectral inequality. Their historical novelty has not been established. The strongest boundaries are the inherited global action/continuation argument, absent spatial stability theorem, and absent certified computation for a nontrivial continuum instance. The applicability score in the catalogue is not used as a contribution score.

A Exact local corpus locators

These are actual passages read and used, not routing-map substitutes. Line numbers are intentionally not treated as stable identifiers.

1. **RamCS.** `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch07-ramcs-rev2.tex`: sections *A centered complex transform*, *Inversion on a specified contour*, *Differentiation and boundary terms*, *Convolution and the moment generator*, *Plane waves and actual eigenfunctions*, *Exact symbol representation*, *Why pointwise reciprocal symbols are not exact inverses*, and *Spectral zeros and stable inversion*; labels `eq:ramcs-definition`, `thm:ramcs-inversion`, `eq:ramcs-plancherel`, `eq:ramcs-convolution`, `eq:variable-plane-wave`. The main chapter was read through *Weighted contours and conditioning*. Also read `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch07-contours-C.tex` in full, especially *Fourier and Laplace restrictions with all signs visible*.

2. **Physical metric.** `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`: full file, particularly *A Nonorthogonal Representation Carries a Metric*, *The Full Residual Has an Omitted Component*, and *Approximating*

the Metric; labels `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, `v21c01:gram-equivalence`.

3. **Checked finite algebra.** `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`: *Polynomial evaluation without eigendecomposition*, *Elimination is not pooling*, *Topology Edits and Checked Low-Rank Reuse*, and the opening *The full updated residual* subsection of *Residuals, Error Ledgers, and Query Certificates*; labels `eq:filtererror`, `prop:schurresidual`, `eq:schurresidual`, `thm:woodbury`, `eq:inexactwoodbury`, `eq:residualbound`.

References

[1] OpenAI, *Sharp one-dimensional Lieb–Thirring inequalities for matrix potentials*, October 5, 2026, source pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; pinned public source directory. Source locators throughout the report are relative to its `build` directory.

[2] A. Laptev and T. Weidl, *Sharp Lieb–Thirring inequalities in high dimensions*, Acta Mathematica **184** (2000), 87–111; arXiv:math-ph/9903007; doi:10.1007/BF02392782.

[3] R. Benguria and M. Loss, *A simple proof of a theorem of Laptev and Weidl*, Mathematical Research Letters **7** (2000), 195–203; arXiv:math-ph/9906024; doi:10.4310/MRL.2000.v7.n2.a5.

[4] L. Read and M. R. Schulz, *The sharp one-dimensional Lieb–Thirring inequality for the sum of eigenvalues*, version 2, September 14, 2026; arXiv:2609.10478v2, Proposition 4 and equation (2.1). Primary online text consulted October 7, 2026.

[5] Confidential Ram Calculus corpus, Volume 1 revision 2, *RamCS Transforms and Operator Spectra*, main chapter and contour supplement; exact local paths and section labels in Appendix A.

[6] Confidential Ram Calculus corpus, Volume 21, *A Nonorthogonal Representation Carries a Metric* and subsequent physical-metric sections; exact local path and labels in Appendix A.

[7] Confidential corpus, *TR-core-pat-5*, finite inverse, Schur, update, and residual passages; exact local path and labels in Appendix A.